Technical Assignment: AWS VPC Terraform Module

Objective

Create a reusable Terraform module that provisions a basic AWS networking environment consisting of a VPC, public subnets, and private subnets.

The goal of this assignment is to demonstrate your ability to write clean, maintainable, and reusable Terraform code while applying fundamental AWS networking concepts.

Requirements

1. VPC

Create an AWS VPC with:

- A configurable CIDR block
- DNS support and DNS hostnames enabled
- Configurable resource tags

2. Subnets

The module should support:

- Creating multiple public subnets
- Creating multiple private subnets
- Configurable CIDR blocks for each subnet
- Configurable availability zones
- Appropriate configuration for public and private subnets

The module should be reusable across different AWS regions and should not assume a fixed number of subnets.

3. Internet Connectivity and Routing

Implement the required networking resources so that:

- Public subnets have a route to the internet through an Internet Gateway
- Private subnets do not have a direct route to the internet
- Public and private subnets use separate route tables
- All subnets are associated with the appropriate route table

- NAT Gateways, NAT instances, and VPC endpoints are explicitly out of scope.

4. Module Interface

Design an appropriate module interface that allows other Terraform configurations to consume your module.

The module should:

- Expose meaningful input variables with appropriate types and descriptions
- Use sensible defaults where appropriate
- Provide useful outputs
- Support consistent tagging of resources

You are free to choose how to structure the input variables and outputs.

5. Example Configuration

Provide an example demonstrating how to use the module to create:

- One VPC with CIDR 10.0.0.0/16
- Two public subnets in different availability zones
- Two private subnets in different availability zones
- Appropriate routing for both subnet types
- Tags identifying the environment and project

The example should demonstrate how the module can be consumed from a separate Terraform configuration.

6. Documentation

Include a README containing:

- A brief description of the module
- Usage instructions
- Input and output documentation
- Any assumptions or design decisions

7. Code Quality

We expect the solution to:

- Follow Terraform best practices and naming conventions
- Specify Terraform and AWS provider version constraints
- Pass terraform fmt and terraform validate
- Avoid unnecessary complexity

Deliverables

Please provide your code in form of a Git repository or a zipped folder. You do not need to deploy the infrastructure to an AWS account. We will review the Terraform code and its design.